

MATLAB Examples

Examples Home > MATLAB Family > Computational Finance > Econometrics Toolbox > Multivariate Models

By MathWorks 

Explore:

[Econometrics Toolbox](#)

Copy and run in your MATLAB:

```
openExample('econ/EstimateVECMoelParametersExample')
```



View in: [Documentation](#)

Estimate VEC Model Parameters Using egctest

This example shows how to estimate the parameters of a vector error-correction (VEC) model. Before estimating VEC model parameters, you must determine whether there are any cointegrating relations (see [docid:econ_ug.buwhxjg-1](#)). You can estimate the remaining VEC model coefficients using ordinary least squares (OLS).

Following from [docid:econ_ug.buwhxjg-1](#), load the Data_Canada data set. Run the Engle-Granger cointegration test on the small-term, medium-term, and long-term interest rate series.

```
load Data_Canada
Y = Data(:,3:end); % Interest rate data
[~,~,~,reg] = egctest(Y,'test','t2');
c0 = reg.coeff(1);
b = reg.coeff(2:3);
beta = [1;-b];
```

Suppose that a model selection procedure indicates the adequacy of $q = 2$ lags in a $VEC(q)$ model. Subsequently, the model is

$$\Delta y_t = \alpha(\beta' y_{t-1} + c_0) + \sum_{i=1}^2 B_i \Delta y_{t-i} + c_1 + \varepsilon_t.$$

Because you estimated c_0 and $\beta = [1; -b]$ previously, you can conditionally estimate α , B_1 , B_2 , and c_1 by:

1. Forming the required lagged differences
2. Regress the first difference of the series onto the q lagged differences and the estimated cointegration term.

Form the lagged difference series.

```
q = 2;
[numObs,numDims] = size(Y);
tBase = (q+2):numObs; % Commensurate time base, all lags
T = length(tBase); % Effective sample size
YLags = lagmatrix(Y,0:(q+1)); % Y(t-k) on observed time base
LY = YLags(tBase,(numDims+1):2*numDims); % Y(t-1) on commensurate time base
```

Form multidimensional differences so that the k^{th} numDims-wide block of columns in DelatYLags contains $(1-L)Y(t-k+1)$.

```

DeltaYLags = zeros(T,(q+1)*numDims);
for k = 1:(q+1)
    DeltaYLags(:,((k-1)*numDims+1):k*numDims) = ...
        YLags(tBase,((k-1)*numDims+1):k*numDims) ...
        - YLags(tBase,(k*numDims+1):(k+1)*numDims);
end

DY = DeltaYLags(:,1:numDims);           % (1-L)Y(t)
DLY = DeltaYLags(:,(numDims+1):end); % [(1-L)Y(t-1),...,(1-L)Y(t-q)]

```

Regress the the first difference of the series onto the q lagged differences and the estimated cointegration term. Include an intercept in the regression.

```

X = [(LY*beta-c0),DLY,ones(T,1)];
P = (X\DY)'; % [alpha,B1,...,Bq,c1]
alpha = P(:,1);
B1 = P(:,2:4);
B2 = P(:,5:7);
c1 = P(:,end);

```

Display the VEC model coefficients.

```
alpha,b,c0,B1,B2,c1
```

$\alpha =$

-0.6336
0.0595
0.0269

$b =$

2.2209
-1.0718

$c_0 =$

-1.2393

$B_1 =$

0.1649	-0.1465	-0.0416
-0.0024	0.3816	-0.3716
0.0815	0.1790	-0.1528

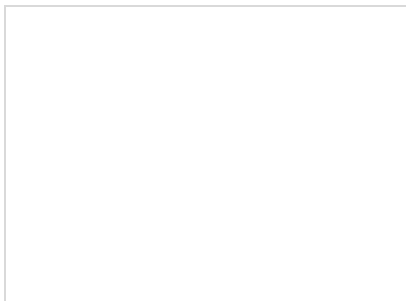
$B_2 =$

-0.3205	0.9506	-0.9514
-0.1996	0.5169	-0.5211
-0.1751	0.6061	-0.5419

$c_1 =$

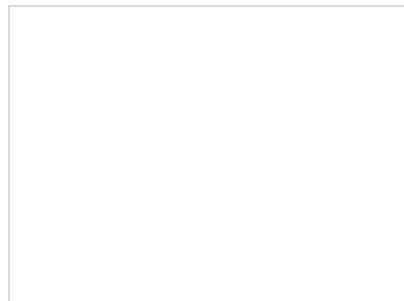
0.1516
0.1508
0.1503

Related Examples



Simulate and Forecast a VEC Model

This example shows how to generate forecasts from a VEC(q) model several ways: Multivariate Models Econometrics Toolbox Computational Finance



Estimate Multiplicative ARIMA Model

This example shows how to estimate a multiplicative seasonal ARIMA model using estimate. The time series is monthly international airline passenger